Ridham Patel

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I am an M.E. Information Technology student at L.D. College of Engineering, Ahmedabad. My research is in machine learning, with work on federated learning (privacy-preserving training and client-level unlearning), computer vision, and the evaluation of deep learning models under real-world conditions. I am looking for research positions in machine learning.

Education

L.D. College of Engineering, Ahmedabad
M.E. Information Technology — *2026 – Present*

- Courses: Advanced Algorithms, Advanced Data Structures, Deep Learning, Mathematics for Machine Learning, Computer Vision, Edge Computing.

LDRP Institute of Technology and Research, Gandhinagar — **CGPA 7.70/10**
B.E. Information Technology — *2022 – 2026*

- Courses: Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Operating Systems, DBMS.
- Qualified GATE 2026 in Data Science & Artificial Intelligence (DA).

Publications

C = Conference J = Journal P = Patent D = Dataset

2026 [C1] **Towards Robust Classroom Attendance: A Comprehensive Evaluation of Face Detection and Recognition Models.** — IEEE CICON, Accepted
H. Trivedi, H. Patel, **R. Patel**, K. Patel, N. Patel

[J1] **FedRevoke: Client-Level Unlearning of Malicious Districts in Lightweight Federated Automatic Number Plate Recognition.** — IEEE OJ-ITS, Under Review
R. Patel, H. Trivedi, C. Dabhi, S. Surati, B. Shrimali

[J2] **FedDETR for Privacy Preserving K Means Based Helmet Detection of Two Wheeler Riders.** — Springer Nature, Under Review
K. Gajjar, N. Patel, **R. Patel**, K. Patel, C. Dabhi, H. Trivedi, M. Goyani, C. Bhatt

[J3] **Limited-Sample Face Recognition through Projection-Efficient Inter-Patch Interaction and Dynamic Patch Reweighting under Real-World Occlusions for Classroom Attendance.** — Elsevier ESWA, Under Review
H. Trivedi, H. Patel, **R. Patel**, K. Patel, N. Patel, D. Patel, K. Savaj, P. Malani, J. Patel

[J4] **Aggregate-Constrained Parcel-Level Yield Estimation from Sparse Single-Polarization X-Band SAR Time Series with Statistical Yield Anchors.** — Springer Nature, Under Review
J. Sorathiya†, Y. Sorathiya†, M. Parmar†, K. Patel†, N. Patel†, **R. Patel**†, H. Trivedi

[D1] **EdgeVision Dataset.** — Mendeley Data, 20k+ downloads
K. Gajjar, N. Patel, **R. Patel**, K. Patel, C. Dabhi, H. Trivedi, M. Goyani, C. Bhatt

2025 [P1] **Automated Waste Segregation Dustbin for Educational Institutions.** — Indian Patent, Under Examination, App. 202521111515
N. Patel, K. Gajjar, **R. Patel**, K. Patel, H. Trivedi

† Equal contribution.

Research and Industry Experience

Associate Software Engineer, OpenXcell Technolabs *Aug. 2025 – May 2026*

- Red-teamed client LLM applications with NVIDIA Garak, probing for jailbreaks, prompt injection and PII leakage.
- Built Guardrails AI defences targeting the failure modes found, and checked that they blocked them before deployment.
- Built retrieval-augmented generation pipelines that ground model outputs in client document collections.

Research Intern, IEEE EMBS Pune Chapter *Jun. 2025 – Jul. 2025*

- Compared 10 deep learning architectures (incl. ResNet50, EfficientNetB2, DenseNet121) for EEG-based schizophrenia detection, measuring accuracy and latency.
- Reached 96% accuracy with ensemble stacking; used Grad-CAM to check which signal regions drove predictions against known diagnostic markers.

Software Development Intern, Institute for Plasma Research, Gandhinagar *Aug. 2024 – Nov. 2024*

- Migrated legacy systems to a web platform (Node.js, TypeScript, SQL) and built data pipelines (Pandas, NumPy) for monitoring experimental data.

Selected Projects

Incident Response Commander: An RL Environment for LLM Agents *2026*
Finalist project, Meta × PyTorch × OpenEnv Hackathon 2026.

- Designed an RL environment simulating SRE incident response over a 10-service system, with a reward covering investigation, root-cause identification, remediation, time and safety.
- Evaluated a Llama 3.1 8B agent: scores of 0.70 / 0.62 / 0.56 on easy / medium / hard tasks.

Face-Swap Deepfake Detection *2024*
Finalist project, Smart India Hackathon 2024, for the National Technical Research Organisation (NTRO).

- Benchmarked ResNet-101, XceptionNet, VGG16, InceptionV3, EfficientNet and BehaviorNet on subsets of FaceForensics++, Celeb-DF, DFDC, Google DFD and WLDR.
- Built a custom evaluation dataset and an ensemble of VGG16, EfficientNetB3 and InceptionV3 to select the final pipeline, integrated into an end-to-end platform for face-swap deepfake detection and report generation.

Skills and Certifications

Programming: Python, TypeScript, SQL, LaTeX.
ML and Data: PyTorch, TensorFlow, scikit-learn, OpenCV, Hugging Face, Pandas, NumPy.
LLMs: LangChain, LangGraph, retrieval-augmented generation (RAG), vector databases.
Systems: Docker, Kubernetes, FastAPI, PostgreSQL, AWS, Git.
Certifications: AWS Certified Cloud Practitioner (CLF-C02).